

JIWANI, Pakistan

WMO: 417560

Lat: 25.068N

Lon: 61.805E

Elev: 190

StdP: 14.60

Time Zone: 5.00 (E05)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.9	57.1	14.3	11.5	69.8	23.8	18.2	68.8	19.8	66.7	16.9	71.6	3.6	0	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.6	98.3	76.1	95.9	76.6	93.9	77.6	83.9	90.5	83.1	89.5	82.5	88.9	7.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.5	170.6	87.4	81.7	165.9	87.1	80.9	161.8	86.8	48.5	90.8	47.6	89.4	46.8	87.7	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	13.8	11.8	48.7	107.9	4.9	5.7	45.2	112.0	42.3	115.4	39.5	118.6	35.9	122.7		
(5)			38.4	85.1	4.4	1.3	35.3	86.0	32.7	86.8	30.3	87.5	27.1	88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	79.8	68.0	70.1	76.1	82.9	87.3	88.1	85.9	84.1	83.8	83.2	77.2	70.7	(6)
(7)		DBStd	7.45	3.57	4.28	3.82	3.25	2.97	2.55	2.18	1.64	1.95	2.38	3.40	4.15	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	36	17	10	0	0	0	0	0	0	0	0	0	9	(9)
(10)		CDD50	10883	558	564	809	986	1155	1142	1112	1056	1014	1030	817	640	(10)
(11)		CDD65	5445	110	154	344	536	690	692	647	591	564	565	367	185	(11)
(12)		CDH74	61743	377	658	2494	5992	9501	9873	8607	7179	6686	6325	3054	997	(12)
(13)		CDH80	27048	11	67	643	2555	5119	5569	4158	2798	2582	2560	884	102	(13)
(14)	Wind	WSAvg	5.0	4.5	4.8	5.2	5.9	5.8	5.8	6.1	5.5	4.3	3.9	4.0	4.4	(14)
(15)	Precipitation	PrecAvg	3.2	0.8	0.4	0.6	0.2	0.0	0.1	0.2	0.0	0.0	0.0	0.2	0.7	(15)
(16)		PrecMax	9.7	7.8	3.5	4.7	2.9	0.0	3.6	3.9	0.5	0.3	0.2	1.9	4.7	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.2	1.3	0.7	1.1	0.6	0.0	0.5	0.7	0.1	0.0	0.0	0.4	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.0	85.7	93.1	97.1	104.3	103.7	95.8	92.3	95.9	97.8	91.5	85.8	(19)
MCWB			62.6	67.1	66.1	70.2	75.8	76.5	82.2	81.5	72.1	69.9	65.3	62.1	(20)	
2%		DB	78.8	81.8	88.4	94.6	98.9	97.9	92.8	90.0	91.8	94.0	88.5	82.4	(21)	
		MCWB	65.0	67.1	67.6	71.5	76.1	80.1	82.0	79.9	77.3	71.1	67.9	65.7	(22)	
5%		DB	77.4	79.6	85.7	92.4	95.9	95.1	91.4	88.9	89.9	91.4	86.5	80.7	(23)	
		MCWB	64.8	66.5	68.8	72.3	76.8	81.3	81.3	79.2	78.1	72.9	69.0	65.9	(24)	
10%		DB	76.1	77.8	83.3	90.1	93.7	93.0	90.1	87.9	88.2	89.3	84.7	79.1	(25)	
		MCWB	64.2	65.9	69.2	73.0	77.9	81.6	80.7	78.7	77.9	74.8	68.9	65.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.6	74.9	76.8	81.0	84.1	85.4	84.4	83.1	81.9	82.0	77.7	74.3	(27)
MCD B			77.9	80.3	83.3	85.9	89.5	91.8	91.0	89.6	87.1	87.9	83.3	79.6	(28)	
2%		WB	69.7	72.2	74.8	79.9	82.8	84.4	83.3	81.7	80.9	80.8	76.3	72.0	(29)	
		MCD B	76.2	77.6	80.8	85.5	88.8	90.9	90.2	87.4	86.5	86.2	82.3	77.7	(30)	
5%		WB	68.3	70.9	73.7	78.7	82.2	83.6	82.5	80.7	80.1	79.9	75.1	70.7	(31)	
		MCD B	74.9	76.5	79.9	84.6	88.5	90.0	89.2	86.7	85.9	85.2	81.6	76.9	(32)	
10%		WB	67.0	69.4	72.8	77.5	81.5	82.8	81.8	79.9	79.5	79.0	74.0	69.3	(33)	
		MCD B	73.5	75.2	79.1	83.7	87.9	89.2	88.5	86.0	85.3	84.4	80.4	76.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.9	14.7	14.4	14.6	13.2	10.6	8.0	7.8	10.1	13.9	15.6	16.6	(35)
MCDBR			16.8	16.0	17.1	17.4	17.6	14.2	9.5	9.0	12.5	17.9	16.8	17.5	(36)	
MCWBR			9.7	7.9	7.6	7.1	6.6	4.9	3.3	3.2	4.3	8.4	8.2	9.9	(37)	
5% WB		MCDBR	15.8	14.4	13.8	14.2	13.5	11.7	9.3	9.0	10.8	13.6	14.4	14.8	(38)	
		MCWBR	10.2	8.1	7.0	7.0	5.2	4.4	3.6	3.9	4.0	5.9	6.8	9.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.405	0.452	0.504	0.550	0.603	0.688	0.757	0.685	0.591	0.498	0.430	0.410	(40)	
taud		2.226	2.081	1.935	1.857	1.772	1.639	1.557	1.672	1.810	1.972	2.151	2.191	(41)		
Ebn at Noon		266	261	253	245	231	211	197	211	228	243	255	258	(42)		
Edh at Noon		40	49	60	66	72	82	88	79	67	54	43	40	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1379	1644	1967	2220	2366	2263	1971	1902	1919	1782	1487	1310	(44)	
(45)		RadStd	58	68	78	70	38	106	118	105	69	42	43	57	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page